

MacOS Hadoop Installation Guide

What is Hadoop?

The Apache Hadoop software library is a framework designed for the distributed processing of large datasets across computer clusters using straightforward programming models. It is built to scale from a single server to thousands of machines, each contributing local computation and storage. Instead of depending on hardware for high availability, Hadoop is engineered to detect and manage failures at the application level, ensuring a reliable and highly available service, even when individual machines within the cluster are susceptible to failures.

Why Install Hadoop on macOS?

- **Local Development and Testing:** Installing Hadoop on macOS empowers developers to build, test, and experiment with big data applications locally before deploying them to larger clusters. This saves time and resources by enabling efficient software development, debugging, and learning without relying on dedicated servers.
- **Unix-Based Compatibility:** macOS's Unix-like foundation closely aligns with Linux, the primary operating system used in large-scale Hadoop clusters. This compatibility makes macOS an ideal platform for installing and running Hadoop, as many commands and configurations are similar across both systems.
- **Accessible studying Environment:** For data engineers, students, and big data aficionados, installing Hadoop on macOS creates a flexible and accessible environment for studying Hadoop's fundamental ideas. Users can get hands-on experience with HDFS, MapReduce, and YARN without relying on cloud platforms or dedicated infrastructure.
- **Harnessing macOS Hardware:** Modern Mac computers, which are equipped with powerful processors, abundant memory, and sufficient storage, can perform small to medium-sized data processing workloads. This qualifies them for Hadoop's resource-intensive operations during development and testing.
- **Cost-Effective Setup:** Individuals and small teams can save money by running Hadoop on a single macOS machine rather than paying for cloud services or several physical servers. This cost-effective strategy enables the exploration of big data technologies without the need to set up a full cluster.

What is Homebrew?

Homebrew is a popular package manager for macOS and Linux systems. Despite its name, the actual command used is simply `brew`. What's interesting about Homebrew is that its commands and options follow a "brewing" theme. In this tutorial, we mention how to install Hadoop with Homebrew. Use the following command to install Homebrew if it is not already in your system.

Check your homebrew installation:

```
brew
```

```
sushil@Sushils-MacBook-Pro ~ % brew
Example usage:
  brew search TEXT|/REGEX/
  brew info [FORMULA|CASK...]
  brew install FORMULA|CASK...
  brew update
  brew upgrade [FORMULA|CASK...]
  brew uninstall FORMULA|CASK...
  brew list [FORMULA|CASK...]
```

Command to install Homebrew: (If not installed already)

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Hadoop Installation Prerequisites:

- **Java Development Kit (JDK):** You know that Hadoop is developed in Java, hence it requires JDK 8 or above. You can download it from Oracle's official website or install OpenJDK with Homebrew using this command:

```
brew update
```

```
brew install openjdk
```

Run this command to create a symbolic link.

```
sudo ln -sf /opt/homebrew/opt/openjdk/libexec/openjdk.jdk  
/Library/Java/JavaVirtualMachines/openjdk.jdk
```

Check java version

```
java -version
```

```
sushil@Sushils-MacBook-Pro ~ % java --version  
openjdk 25.0.2 2026-01-20  
OpenJDK Runtime Environment Homebrew (build 25.0.2)  
OpenJDK 64-Bit Server VM Homebrew (build 25.0.2, mixed mode, sharing)  
sushil@Sushils-MacBook-Pro ~ %
```

Install Hadoop

Now, use Homebrew to fetch the latest stable version of Hadoop.

```
brew install hadoop
```

```
sushil@Sushils-MacBook-Pro ~ % brew install hadoop  
✓ JSON API cask.jws.json Downloaded 15.4MB/ 15.4MB  
✓ JSON API formula.jws.json Downloaded 32.0MB/ 32.0MB  
==> Fetching downloads for: hadoop  
✓ Bottle Manifest hadoop (3.4.3) Downloaded 11.0KB/ 11.0KB  
✓ Bottle Manifest openjdk@11 (11.0.30) Downloaded 40.0KB/ 40.0KB  
✓ Bottle openjdk@11 (11.0.30) Downloaded 187.0MB/187.0MB  
✓ Bottle hadoop (3.4.3) Downloaded 446.4MB/446.4MB  
==> Installing hadoop dependency: openjdk@11  
=> Pouring openjdk@11--11.0.30.arm64_tahoe.bottle.1.tar.gz  
📦 /opt/homebrew/Cellar/openjdk@11/11.0.30: 667 files, 310.4MB  
=> Pouring hadoop--3.4.3.all.bottle.tar.gz  
📦 /opt/homebrew/Cellar/hadoop/3.4.3: 20,228 files, 898.3MB  
=> Running `brew cleanup hadoop`...  
Disable this behaviour by setting `HOMEBREW_NO_INSTALL_CLEANUP=1`.  
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).  
sushil@Sushils-MacBook-Pro ~ %
```

Setup Environment Variables

Homebrew installs software in different locations depending on your Mac's chip. We will use dynamic commands to set the correct paths automatically.

Open your terminal profile:

```
nano ~/.zshrc
```

Add this block to the bottom of the file (notice we are explicitly setting JAVA_HOME here):

```
# --- Homebrew Hadoop Environment Variables ---
export JAVA_HOME=$(/usr/libexec/java_home -v 11)
export HADOOP_HOME="$(brew --prefix hadoop)/libexec"
export HADOOP_CONF_DIR="$HADOOP_HOME/etc/hadoop"
export PATH="$HADOOP_HOME/bin:$HADOOP_HOME/sbin:$PATH"
```

Save and Exit (Ctrl + O, Enter, Ctrl + X). Then apply the changes:

```
source ~/.zshrc
hash -r
```

Check Hadoop installation

```
hadoop version
```

```
sushil@Sushils-MacBook-Pro hadoop % hadoop version
Hadoop 3.4.3
Source code repository https://github.com/apache/hadoop.git -r 9d50c6884666e794e45102260a4017bb31802e1b
Compiled by stevel on 2026-02-13T14:23Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 2331238c4c2929e66645316a32a8613
This command was run using /opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/hadoop-common-3.4.3.jar
```

Configure the Cluster

By default, Homebrew installs Hadoop in “Standalone” mode. To simulate a real cluster, you must edit four XML files located in your Homebrew Hadoop directory. To easily open the folder containing these files, run:

```
cd /opt/homebrew/Cellar/hadoop/3.4.3/libexec/etc/hadoop
```

Edit the following files (e.g., `nano core-site.xml`) and place the properties inside the `<configuration>` tags:

core-site.xml

```
nano core-site.xml
```

Copy-paste the following inside the `configurations` tags inside `<configuration>` tags:

```
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://localhost:9000</value>
</property>
```

Save and Exit (Ctrl + O, Enter, Ctrl + X).

mapred-site.xml

```
nano mapred-site.xml
```

Copy-paste the following inside the `configurations` tags inside `<configuration>` tags:

```
<property>
  <name>mapreduce.framework.name</name>
  <value>local</value>
</property>
<property>
  <name>mapreduce.application.classpath</name>
  <value>$HADOOP_HOME/share/hadoop/mapreduce/*:$HADOOP_HOME/share/hadoop/map
reduce/lib/*</value>
</property>
```

Save and Exit (Ctrl + O, Enter, Ctrl + X).

hdfs-site.xml

```
nano hdfs-site.xml
```

Copy-paste the following inside the configurations tags inside <configuration> tags:

```
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
```

Save and Exit (Ctrl + O, Enter, Ctrl + X).

yarn-site.xml

```
nano yarn-site.xml
```

Copy-paste the following inside the configurations tags inside <configuration> tags:

```
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
```

Save and Exit (Ctrl + O, Enter, Ctrl + X).

Format Remote Login (SSH)

Enable Remote Login

```
sudo systemsetup -setremotelogin on
```

Check Remote Login

```
sudo systemsetup -getremotelogin
```

Hadoop requires passwordless SSH access to localhost to start its worker daemons.

Generates ssh key:

```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
```

Copies the key to your authorized list:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

Sets the correct security permissions

```
chmod 0600 ~/.ssh/authorized_keys
```

Test it by running `ssh localhost`. If it doesn't ask for a password, type exit and proceed. (You may need to enable "Remote Login" in your Mac's System Settings > General > Sharing.)

Format HDFS and Start

Format the file system (Do this only once):

```
hdfs namenode -format
```

Start your cluster:

```
start-all.sh
```

Verify with jps.

Jps

```
sushil@Sushils-MacBook-Pro ~ % jps
29505 ResourceManager
29315 SecondaryNameNode
29607 NodeManager
29177 DataNode
30841 Jps
29070 NameNode
```

You can access the UI at:

<http://localhost:9870>.

Run the Sample MapReduce Job

Create the Input Directory in HDFS

Hadoop expects every user to have their own workspace in HDFS. This command dynamically creates a folder based on your Mac username:

```
hdfs dfs -mkdir -p /user/$USER/input
```

```
sushil@Sushils-MacBook-Pro ~ % hdfs dfs -mkdir -p /user/$USER/input
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/hadoop-common-3.4.3.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled
2026-04-05 12:15:32,277 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
```


Create a Local Text File

```
echo "hello hadoop hello macbook homebrew hadoop test" >
~/wordcount_test.txt
```

```
sushil@Sushils-MacBook-Pro ~ % echo "hello hadoop hello macbook homebrew hadoop test" > ~/wordcount_test.txt
sushil@Sushils-MacBook-Pro ~ %
```

Upload the File to HDFS

```
hdfs dfs -put ~/wordcount_test.txt /user/$USER/input/
```

```
sushil@Sushils-MacBook-Pro ~ % hdfs dfs -put ~/wordcount_test.txt /user/$USER/input/
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/hadoop-common-3.4.3.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-04-05 12:16:59,372 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
```

Delete the Old File if it is there.

```
hdfs dfs -rm -r /user/$USER/output
```

Run the MapReduce Job

```
hadoop jar
$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar
wordcount /user/$USER/input /user/$USER/output
```

```
sushil@sushils-MacBook-Pro ~ % hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /user/$USER/input /user/$USER/output

WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/hadoop-common-3.4.3.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-04-05 12:18:22,290 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/user/sushil/output already exists
    at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
    at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:278)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:142)
```

View the Results

```
hdfs dfs -cat /user/$USER/output/part-r-00000
```

```
sushil@sushils-MacBook-Pro ~ % hdfs dfs -cat /user/$USER/output/part-r-00000

WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by org.apache.hadoop.util.NativeCodeLoader in an unnamed module (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/hadoop-common-3.4.3.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-04-05 12:26:05,183 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor (file:/opt/homebrew/Cellar/hadoop/3.4.3/libexec/share/hadoop/common/lib/hadoop-shaded-protobuf_3_25-1.5.0.jar)
WARNING: Please consider reporting this to the maintainers of class org.apache.hadoop.thirdparty.protobuf.UnsafeUtil$MemoryAccessor
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
hadoop      2
hello       2
homebrew    1
macbook     1
test        1
```